

Children's National Hospital

Comment on HHS Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Cover Letter

Children's National Hospital
111 Michigan Avenue NW
Washington, DC 20010

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Docket: HHS Health Sector AI RFI (RIN 0955-AA13)

Dear Sir or Madam,

Children's National Hospital appreciates the opportunity to respond to the Department of Health and Human Services' Request for Information on accelerating the adoption and use of artificial intelligence (AI) as part of clinical care.

As one of the nation's leading pediatric academic medical centers and a pioneer in the development of machine learning technology for children's health, Children's National offers a distinct perspective on AI adoption. Caring for minors introduces heightened considerations related to privacy, consent and assent, data representativeness, safety thresholds, and family trust.

Our comments focus on regulatory clarity, pediatric AI infrastructure, liability and accountability, and implementation science. Our team has expertise in guiding safe and trusted AI development through its lifecycle, from ideation and design to clinical implementation and adoption. We view this RFI as the beginning of an ongoing dialogue and welcome continued engagement.

Respectfully submitted,
Children's National Hospital

Regulations.gov Comment

This submission reflects Children's National Hospital's response to the HHS Request for Information on Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care. The response emphasizes a pediatric-focused lens, risk-proportionate governance, national pediatric AI infrastructure, and shared accountability models to enable safe and responsible AI adoption.

Children's National Hospital Comment on

HHS Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Executive Summary

Children's National Hospital strongly supports HHS's efforts to accelerate the responsible adoption of artificial intelligence in clinical care. We submit these comments to emphasize the importance of a **pediatric-first policy lens**. AI used in the care of minors raises distinct considerations related to privacy, consent, data sufficiency, clinical risk tolerance, and family trust that are not fully addressed by adult-centric frameworks.

We recommend that HHS prioritize:

1. Risk proportionate regulation of AI technology with mandatory transparency around pediatric performance and training data.
2. National pediatric AI research collaboration and governance infrastructure, including federated data and model evaluation resources.
3. Multi-stakeholder approaches to liability and accountability that enable trust and adoption without negatively impacting innovation.
4. Implementation science, clinician and family education, and pilots aligned to pediatric outcomes and burden reduction.
5. Establishment of a national pediatric AI council, structured as a public-private partnership to provide coordinated guidance on standards, evaluation, deployment, and oversight of AI systems affecting children.

I. Regulation

Pediatric-Specific Transparency and Algorithmic Accountability

For both medical and non-medical device AI used in clinical care, Children's National recommends that HHS establish clear expectations for **algorithmic accountability**, including standardized disclosures by AI developers and vendors. These disclosures should include:

- The populations on which models were trained and validated, with explicit identification of pediatric representation and appropriate stratification.
- Transparent performance limitations and failure modes, particularly where pediatric data are sparse or absent.
- Ongoing monitoring and update strategies for models that learn or evolve over time.

In pediatrics, the absence or under-representation of children in training data can materially affect safety and effectiveness. Transparency regarding training populations would allow health systems to make informed adoption decisions and protect patients.

Privacy, Consent, and Behavioral Health

Pediatric care introduces additional complexity related to guardianship, adolescent confidentiality, and sensitive behavioral health data. HHS guidance clarifying how existing privacy and consent frameworks apply when AI systems process or generate clinical content would significantly reduce uncertainty and variability across organizations.

Scalable Governance Models

Only a small number of pediatric institutions currently have the resources to implement comprehensive AI governance. HHS can accelerate safe adoption by supporting the development of **shared pediatric governance frameworks**, policies, and monitoring tools that can be guided by major clinical centers equipped for AI deployment and broadly adopted across children's hospitals.

II. Reimbursement

Children's National recognizes that reimbursement policy for AI remains an evolving area. We encourage HHS to explore **pilot programs** that align payment with measurable pediatric value, including:

- Reductions in avoidable emergency department utilization and hospitalizations.
- Improvements in preventive care adherence.
- Clinician time savings from AI-enabled documentation and workflow tools, allowing greater focus on patient and family engagement.
- Cost savings from AI-enabled early diagnosis and treatment.

Such pilots should be paired with rigorous evaluation to inform longer term policy decisions.

III. Research and Development

A National Pediatric AI Consortium

Children's National strongly supports the creation of a national, HHS led **Pediatric AI Consortium**. Pediatric cohorts are often small at individual institutions, limiting model

development and evaluation. A federated approach—enabling shared learning while preserving privacy—would:

- Expand access to high quality pediatric AI resources.
- Support benchmarking, robustness testing, and bias assessment.
- Enable participation by smaller and under-resourced pediatric centers.

Under HHS’s leadership, the consortium could create and manage national public resources for pediatric AI including open-access datasets, AI benchmarking initiatives, and publicly maintained code repositories--to support transparency, reproducibility, and continued development.

Methodological Focus and Rare Diseases

One in three children born with a rare disease—a group of conditions that collectively affect 25–30 million Americans—will not survive to their fifth birthday. We recommend prioritizing **AI methodologies suitable for data-limited conditions**, including rare diseases that impact the survival and quality of life of patients of all ages. Investments in methods such as federated learning, data-efficient modeling, and computationally efficient deployment will have broad downstream impact.

Predictive and Preventive Care

Early detection of health risks, coupled with timely intervention, can save lives and improve quality of life across the lifespan. We recommend prioritizing **predictive AI methodologies** that integrate multimodal information—including electronic health records, clinical notes, lab results, imaging studies, and patient history—to identify risks prior to symptom onset. Such approaches can promote and support preventive care strategies that enable children at risk to grow into healthy adults.

Implementation Science and Public Education

Patient and caregiver acceptance remains a significant barrier to AI adoption in pediatrics. HHS support for implementation science, clinician training, and public education initiatives will be essential to building trust and understanding of AI’s appropriate role in care.

IV. Responses to Specific HHS Questions

1. What are the biggest barriers to private-sector innovation in AI for health care and its adoption and use in clinical care?

From a pediatric health system perspective, the most significant barriers to AI adoption and innovation include:

Limited pediatric data availability and fragmentation. Pediatric cohorts at individual institutions are relatively small, particularly for rare diseases, neonatal populations, and subspecialty conditions. While data sharing consortia exist, few are oriented toward the full lifecycle needs of AI development, validation, and deployment.

Resource and capability constraints. Only a small number of pediatric institutions possess the technical, operational, and governance capacity to responsibly develop or adopt AI. This creates barriers in access and slows system-wide learning.

Lack of pediatric-specific performance transparency. Many AI tools entering the market lack clear disclosure regarding the populations on which they were trained or validated. In pediatrics, the absence or under-representation of children in training data can materially affect safety and effectiveness.

Legal and liability uncertainty. The absence of clear accountability frameworks, particularly for non-medical-device AI, creates risk aversion among clinicians and health systems, directly impeding adoption.

Trust and acceptance gaps. While clinicians are becoming more receptive to AI as they gain experience, patient and caregiver trust remains uneven, underscoring the need for broad education and transparency across all AI users and stakeholders.

HHS can accelerate adoption by reducing pediatric data fragmentation, investing in shared AI infrastructure, and lowering resource and governance barriers so that innovation is not limited to a small number of well-resourced pediatric institutions.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize effective AI use in clinical care?

Children's National recommends that HHS prioritize the following:

Risk-proportionate governance for clinical AI. HHS should establish clear expectations for AI systems used in clinical care that scale oversight to clinical risk, autonomy, and impact—without treating all AI as regulated medical devices.

Mandatory transparency and disclosure expectations. AI vendors should be required to disclose training populations, intended use, known limitations, and pediatric performance metrics where applicable. This would improve informed decision-making and patient safety.

Programmatic support for pediatric AI infrastructure. Federal investment in shared pediatric AI resources—benchmarks, federated validation environments, and governance templates—would accelerate adoption across diverse pediatric care settings.

Exploratory payment pilots. While pediatric reimbursement remains complex, HHS should pilot outcome-aligned reimbursement models that recognize AI’s contributions to care quality, clinician efficiency, and caregiver burden reduction.

HHS can accelerate adoption by establishing risk-proportionate expectations for medical-device and non-medical-device AI, requiring pediatric performance transparency, and piloting payment models aligned to measurable pediatric value.

3. For non-medical devices, what novel legal and implementation issues exist, and what role should HHS play?

Liability and accountability for non-medical-device AI represent one of the most significant unresolved barriers to adoption.

Currently, accountability largely rests with the individual clinician using AI, even when the underlying model is opaque, continuously learning, or externally developed. This dynamic undermines efficiency gains and discourages responsible adoption.

Children’s National recommends that HHS:

- Convene a multi-stakeholder working group including clinicians, health systems, AI vendors, regulators, payers, and insurers.
- Examine shared-responsibility models that align accountability with control and transparency.
- Explore the role of insurance and contractual safe harbors for organizations that meet defined governance and monitoring standards.

HHS is uniquely positioned to facilitate consensus in this area and reduce fragmentation across the health sector. **HHS can accelerate adoption** by convening multi-stakeholder efforts to develop shared accountability and liability frameworks that align responsibility with transparency, control, and governance maturity.

4. For non-medical devices, what are the most promising AI evaluation methods, and should HHS further support these processes?

In pediatrics, effective evaluation must account for age-related variation, developmental differences, and smaller cohorts.

Promising approaches include:

- Pre-deployment validation using pediatric-specific datasets and age-stratified metrics.
- Prospective monitoring in “shadow mode” prior to clinical activation.
- Ongoing post-deployment surveillance for model drift, bias, and alert burden.
- Human-centered evaluation assessing workflow integration and cognitive load.

HHS support for standardized pediatric evaluation frameworks, open benchmarking tools, and applied research funding would substantially improve consistency and safety. **HHS can accelerate adoption** by supporting standardized pediatric evaluation frameworks, age-stratified performance benchmarks, and post-deployment monitoring resources applicable across diverse care settings.

5. How can HHS support private-sector activities such as accreditation, certification, and testing?

HHS can support voluntary, industry-driven approaches by:

- Endorsing standardized AI “model cards” tailored to clinical and pediatric contexts.
- Supporting public registries that track AI tools, validated use cases, and reported safety events.
- Encouraging harmonization of testing and certification efforts to reduce duplicative burden.

Such efforts should prioritize transparency and learning over prescriptive regulation. **HHS can accelerate adoption** by endorsing voluntary, harmonized transparency and certification mechanisms that prioritize learning, safety, and consistency without imposing unnecessary regulatory burden.

6. Where have AI tools deployed in clinical care met or exceeded expectations, and where have they fallen short?

Areas exceeding expectations include operational and administrative domains such as clinical documentation, summarization, and workflow optimization. These applications reduce burden without directly introducing patient safety risk. In clinical space, the earliest and most successful adoption is in radiology and imaging sciences where decisions are complex and clinical burnout is high.

Areas falling short include highly specialized, high-stakes clinical decision-making with imminent impact on a patient's health, where decision accountability is undefined. Particularly in pediatrics, trust, data sufficiency, and validation remain challenges. The lack of explainability in the process of decision of certain AI technologies negatively impacts adoption of the trust of clinical users.

High-potential future applications include care coordination copilots, medication reconciliation for complex pediatric dosing, rare-disease pattern detection, and AI-assisted imaging workflows that support, but do not replace, clinical judgment.

HHS can accelerate adoption by focusing incentives and research on high-value pediatric use cases while supporting the data, validation, and trust foundations needed for safe expansion into higher-stakes clinical domains as well as by establishing AI-and-human-in-the-loop clinical workflows with defined and structured accountability of the decision processes.

7. Which roles and governing bodies most influence AI adoption within health care organizations?

In pediatric institutions, AI adoption is shaped by a combination of clinical leadership, nursing leadership, informatics, technical leadership, research leadership, privacy and risk management, and legal and compliance teams. Administrative hurdles include procurement complexity, EHR integration, data accessibility, and workforce education.

HHS can accelerate adoption by providing shared policy templates, governance standards, and implementation guides tailored to pediatric settings.

8. Where would enhanced interoperability widen market opportunities and accelerate AI development?

Key pediatric data domains include growth trajectories, immunization records, neonatal monitoring data, weight-based medication dosing, and behavioral health information. Enhanced interoperability through pediatric-specific standards and federated access models would improve both market competition and research productivity.

HHS can accelerate adoption by advancing pediatric-specific data standards and federated interoperability models that enable scalable AI development while preserving privacy and institutional autonomy.

9. What challenges do patients and caregivers wish to see addressed, and what concerns do they have?

Patients and caregivers seek improved care navigation, reduced administrative burden, and earlier detection of health issues. At the same time, concerns remain regarding privacy, misuse of sensitive data, and over-reliance on automated decision-making.

Transparent communication, clear consent options, and education are essential to addressing these concerns.

HHS can accelerate adoption by investing in transparency, education, and consent-aware AI practices that build trust among patients, caregivers, and families, particularly in sensitive pediatric contexts.

10. Are there specific areas of AI research HHS should prioritize?

Children's National recommends prioritizing:

- Longitudinal studies of AI's impact on pediatric outcomes and equity.
- Research on clinician-AI teaming and workflow optimization.
- Economic evaluations that incorporate caregiver burden and social impact.
- Methods research addressing data scarcity, rare diseases, and bias mitigation.

HHS can accelerate adoption by prioritizing pediatric AI research that addresses data scarcity, longitudinal outcomes, caregiver burden, and real-world implementation effectiveness.

V. Closing

For AI technologies to be adopted broadly and efficiently across national clinical settings, healthcare institutions must address a set of clinical, cultural, computational, and regulatory considerations that have not previously been encountered at scale. At Children's National Hospital, our team has contributed to the development of practical recommendations for AI deployment, emphasizing the need to align clinical staffing, data workflows, and computational infrastructure to enable safe implementation, monitoring, and continuous evaluation of AI systems. In parallel, Children's National Hospital plays leadership roles in the professional societies most influential in both the development and adoption of clinical AI, including the Medical Image Computing and Computer Assisted Intervention Society (MICCAI), the Radiological Society of North America (RSNA), and the American Academy of Pediatrics (AAP).

We stand ready to support HHS' initiatives to advance responsible AI adoption and recommend the establishment of a **National Pediatric AI Council**, structured as a public-private partnership of scientists, clinicians, ethicists, patient and family representatives, regulators, and industry stakeholders. Such a council would provide coordinated guidance on standards, evaluation, deployment, and oversight of AI systems affecting children.

The pediatric healthcare community has reached a pivotal moment for improving their patient well-being. AI has the potential to deliver significant gains in efficiency, access, and quality across the national healthcare system if it is deployed safely and adopted responsibly. Children's National Hospital appreciates HHS's leadership in this area and welcomes

continued engagement on policies and programs that will enable AI to improve pediatric health outcomes nationwide.